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Collected Mathematical Papers

Volume IX, Pages 529–540 (Papers 617–619)

Transcribed from the Original Publications

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617. On the Scalene Transformation of a Plane Curve (continued)

[p. 529] and the transformed curve is

$$r^2 + 2\gamma r\cos\theta + m^2 - n^2 = 0,$$

or, as this may be written,

$$r^2 + 2\gamma r\cos\theta + \gamma^2 - f^2 = 0,$$

where $\gamma^2 - f^2 = m^2 - n^2$, that is, $f^2 + m^2 = \gamma^2 + n^2$; viz. this is a concentric circle radius f.

The theorem may be presented as follows. Consider two concentric circles, centre O and radii h, f respectively; take an arbitrary point B, distance $OB = \gamma$; and taking m arbitrary, determine l, n by the equations

$$l^2 = m^2 + h^2 - \gamma^2, \quad n^2 = m^2 + f^2 - \gamma^2;$$

then drawing through B an arbitrary line to meet the circles in A, C respectively; also describing a circle, centre B and radius $= m$; and through A drawing a line ABC perpendicular to BA to meet the last-mentioned circle in two points P, Q: for these points, the distances from the points A, B, C are $= l$, m, n respectively.

To verify this, take O as the origin, OB for the axis of x, θ the inclination of ABC to this axis, $BA = r'$, $BC = r$; the coordinates of C, B, A are

$$\gamma + r\cos\theta, \quad r\sin\theta,$$

$$\gamma, \quad 0,$$

$$\gamma + r'\cos\theta, \quad -r'\sin\theta,$$

whence, taking (x, y) for the coordinates of P (or Q), the equations to be verified are

$$(x - \gamma - r\cos\theta)^2 + (y - r\sin\theta)^2 = n^2,$$

$$(x - \gamma)^2 + y^2 = m^2,$$

$$(x - \gamma + r'\cos\theta)^2 + (y + r'\sin\theta)^2 = l^2.$$

By means of the second equation, the other two become

$$-2(x - \gamma)r\cos\theta - 2yr\sin\theta + r^2 = n^2 - m^2,$$

$$2(x - \gamma)r'\cos\theta + 2yr'\sin\theta + r'^2 = l^2 - m^2;$$

or, substituting for $n^2 - m^2$, $l^2 - m^2$ the values $f^2 - \gamma^2$ and $h^2 - \gamma^2$, the equations are

$$-2xr\cos\theta - 2yr\sin\theta + 2\gamma r\cos\theta + r^2 - f^2 + \gamma^2 = 0,$$

$$2xr'\cos\theta + 2yr'\sin\theta - 2\gamma r'\cos\theta + r'^2 - h^2 + \gamma^2 = 0,$$

viz. in virtue of the equations of the two circles, these reduce themselves each of them to

$$x\cos\theta + y\sin\theta = 0,$$

which equation, together with the second equation

$$(x - \gamma)^2 + y^2 = m^2,$$

determine (x, y) as above.

[p. 530] Reverting to the case where the locus of A is the circle

$$r'^2 - 2\gamma r' \cos\theta + \gamma^2 - h^2 = 0,$$

this gives

$$r' = \gamma\cos\theta + \sqrt{(h^2 - \gamma^2\sin^2\theta)},$$

$$\frac{1}{r'} = \frac{\gamma\cos\theta - \sqrt{(h^2 - \gamma^2\sin^2\theta)}}{\gamma^2 - h^2},$$

so that for the transformed curve we have

$$r^2 + \gamma\cos\theta\left\{(1-\lambda)r + (1+\lambda)\frac{\gamma^2 - h^2}{r}\right\} + (1-\lambda)\sqrt{(h^2 - \gamma^2\sin^2\theta)} + m^2 - n^2 = 0.$$

Putting for shortness $\dfrac{l^2 - m^2}{\gamma^2 - h^2} = \lambda$, and for r, $r\cos\theta$, $r\sin\theta$, writing $\sqrt{(x^2+y^2)}$, x, y respectively, this is

$$x^2 + y^2 + (1-\lambda)\gamma x + (1+\lambda)\sqrt{[h^2(x^2+y^2) - \gamma^2 y^2]} + m^2 - n^2 = 0,$$

or, what is the same thing,

$$\{x^2 + y^2 + (1-\lambda)\gamma x + m^2 - n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2) - \gamma^2 y^2\},$$

$=$ a bicircular quartic. In the case $\lambda = -1$, it reduces itself to the circle

$$x^2 + y^2 + 2\gamma x + m^2 - n^2 = 0$$

twice, which is the case considered above; and in the case $\lambda = 1$, or $l^2 - m^2 = h^2 - \gamma^2$, the equation is

$$(x^2 + y^2 + m^2 - n^2)^2 = 4\{h^2(x^2+y^2) - \gamma^2 y^2\},$$

so that the curve is symmetrical in regard to each axis. In the case $\gamma = 0$, the locus is a pair of concentric circles, centre B.

The equation

$$\{x^2 + y^2 + (1-\lambda)\gamma x + m^2 - n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2) - \gamma^2 y^2\},$$

which contains the four constants λ, γ, h and $m^2 - n^2$, may be written in the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

(where the constants A, B, a, e are also arbitrary). This is, in fact, the equation of the general symmetrical bicircular quartic, referred to a properly-selected point on the axis as origin, viz. the origin is the centre of any one of the three involutions formed by the vertices (or points on the axis); say it is any one of the three involution-centres of the curve.

To show this, assume

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta) = x^4 - px^3 + qx^2 - rx + s;$$

[p. 531] then, taking B arbitrary, the equation of the symmetrical bicircular quartic having for vertices the points $x = \alpha$, $x = \beta$, $x = \gamma$, $x = \delta$, is

$$(x^2 + y^2 - \tfrac{1}{2}px + B)^2 = (2B + \tfrac{1}{4}p^2 - q)x^2 + (b + 2\theta A)x + (-s + B^2),$$

in fact, this is the form of the general equation, and writing therein $y = 0$, it becomes $x^4 - px^3 + qx^2 - rx + s = 0$, that is, $(x-\alpha)(x-\beta)(x-\gamma)(x-\delta) = 0$. Hence, writing for convenience

$$A = -\tfrac{1}{2}p,$$

$$a = 2B + \tfrac{1}{4}p^2 - q,$$

$$b = r - pB,$$

$$c = -s + B^2,$$

the equation is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + bx + c.$$

This may be written

$$(x^2 + y^2 + Ax + B + \theta)^2 = (a + 2\theta)x^2 + 2\theta y^2 + (b + 2\theta A)x + c + 2B\theta + \theta^2,$$

viz. assuming $\theta = -\dfrac{b}{2A}$ in order on the right-hand side to destroy the term in x, the equation is

$$\left(x^2 + y^2 + Ax + B - \frac{b}{2A}\right)^2 = \left(a - \frac{b}{A}\right)x^2 + y^2 \cdot \frac{-b}{A} + \frac{1}{4A^2}\,(b^2 - 4ABb + 4A^2c),$$

which is of the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2 + f;$$

and if $f = 0$, that is, if $b^2 - 4ABb + 4A^2c = 0$, then it is of the required form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2.$$

We have

$$\begin{aligned} b^2 - 4ABb + 4A^2c &= (r - pB)^2 + 2pB(r - pB) + p^2(-s + B^2) \\ &= r^2 - p^2 s, \end{aligned}$$

or the required condition is $r^2 - p^2 s = 0$. But we have

$$r^2 - p^2 s = (\alpha\delta - \beta\gamma)(\beta\delta - \gamma\alpha)(\gamma\delta - \alpha\beta),$$

as is easily verified by writing

$$p = \delta + p_{\prime}, \quad q = \delta p_{\prime} + q_{\prime}, \quad r = \delta q_{\prime} + r_{\prime}, \quad s = \delta r_{\prime},$$

where $p_{\prime}$, $q_{\prime}$, $r_{\prime}$ stand for

$$\alpha + \beta + \gamma, \quad \beta\gamma + \gamma\alpha + \alpha\beta, \quad \alpha\beta\gamma,$$

[p. 532] respectively. The required condition thus is

$$(\alpha\delta - \beta\gamma)(\beta\delta - \gamma\alpha)(\gamma\delta - \alpha\beta) = 0,$$

viz. the origin (that is, the fixed point B of the cell) must be at one of the three involution-centres.

Comparing the equation

$$\{x^2 + y^2 + (1 - \lambda)\gamma x + m^2 - n^2\}^2 = (1 + \lambda)^2\{h^2(x^2 + y^2) - \gamma^2 y^2\}$$

with the equation

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

we have

$$A = (1-\lambda)\gamma,$$
$$B = m^2 - n^2,$$
$$a = (1+\lambda)^2 h^2,$$
$$e = (1+\lambda)^2 (h^2 - \gamma^2),$$

and thence $a - e = (1+\lambda)^2\gamma^2$. Consequently $\dfrac{a-e}{A^2} = \left(\dfrac{1+\lambda}{1-\lambda}\right)^2$, which gives λ; and then

$$h^2 = \frac{a}{(1+\lambda)^2}, \quad \gamma^2 = \frac{a-e}{(1+\lambda)^2}, \quad m^2 - n^2 = B;$$

viz. we thus have the values of λ, h, γ and $m^2 - n^2$ for the description of a given curve $(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2$. In order that the description may be possible, a and $a - e$ must be each of them positive.

For the Cartesian a is $= e$, whence $1 + \lambda = 0$, and the equation becomes

$$(x^2 + y^2 + 2\gamma x + m^2 - n^2)^2 = 0,$$

which is a twice repeated circle; hence the Cartesian cannot be constructed by means of a cell as above.

To obtain a construction of the Cartesian, it may be remarked that, if a symmetrical bicircular quartic be inverted in regard to an axial focus, viz. if the focus be taken as the centre of inversion, we obtain a Cartesian. The axial foci of the curve

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2$$

are points on the axis, the abscissa $x = \theta$ being determined by the equation

$$e(\theta^2 + A\theta + B)^2 - a(\theta^2 - B)^2 - ae\theta^2 = 0.$$

The equation referred to a focus as origin is therefore

$$\{x^2 + y^2 + (A + 2\theta)x + \theta^2 + A\theta + B\}^2 = ax^2 + ey^2 + 2a\theta x + a\theta^2;$$

then inverting, viz. for x, y writing $\dfrac{k^2 x}{r^2}$, $\dfrac{k^2 y}{r^2}$ (k arbitrary), we have, as may be verified the equation of a Cartesian.

[p. 533] The inversion can be performed mechanically by an ordinary Peaucellier-cell; the complete apparatus for the construction of a Cartesian is therefore as in fig. 2, viz. we have a cell BAC as before, B a fixed point, locus of A a circle (for convenience of drawing, the arrangement has been made BAC instead of ABC), and we connect with C a Peaucellier-cell $CA'B'$, arms n, n', m', the fixed point B' being on the axis, which is the line joining B with the centre of the circle described by A. This being so, then A describing a circle, C will describe a symmetrical bicircular quartic, and A' will describe the inverse of this, being in general a like curve; but if the position of B' be properly determined, viz. if B' be at a focus of the first-mentioned quartic.

Fig. 2.

then A' will describe a Cartesian. A further investigation would be necessary in order to determine how to adapt the apparatus to the description of a given Cartesian.

A more convenient mechanical description of a Cartesian is, however, that given in the paper which follows the present one [618].

The equation

$$\{x^2+y^2+(1-\lambda)\gamma x+m^2-n^2\}^2=(1+\lambda)^2\{h^2(x^2+y^2)-\gamma^2y^2\}$$

may also be written

$$\begin{gathered}\{x^2+y^2+(1-\lambda)\gamma x-\tfrac{1}{2}(1+\lambda)^2(h^2-\gamma^2)+m^2-n^2\}^2\\ =(1+\lambda)^2\{\gamma^2x^2-(1-\lambda)(h^2-\gamma^2)\gamma x-\tfrac{1}{4}(1+\lambda)^2(h^2-\gamma^2)^2-(m^2-n^2)(h^2-\gamma^2)\},\end{gathered}$$

viz. the equation is now brought into the form

$$(x^2+y^2+Ax+B)^2=ax^2+bx+c.$$

Expressing the coefficients A, B, a, b, c in terms of λ, γ, h, m^2-n^2, it appears by what precedes, that we should have identically $b^2-4ABb+4A^2c=0$, viz. this is the equation which expresses that the origin is an involution-centre.

If, instead of the original cell, we consider a new cell obtained by substituting for the arms PB, BQ, the arms pb, bq, jointed on to the points p, q on the arms CP, CQ respectively, and instead of B, making b the fixed point; then writing $Cp=kn$, $pb=km$, so that the parameters of the cell are l, m, n, k, and taking $Ch=s$, $hA=s'$,

[p. 534] we have $s=kr$, $s+s'=r+r'$, that is, $r=\dfrac{s}{k}$, $r'=\dfrac{k-1}{k}s+s'$. Substituting in the equation between r, r', written for greater convenience in the form

$$(r+r')(rr'+m^2-l^2)+(l^2-n^2)r'=0,$$

the relation between s, s' is found to be

$$(s+s')\left(\frac{k-1}{k^2}s^2+\frac{ss'}{k}+m^2-l^2\right)+(l^2-n^2)\left(\frac{k-1}{k}s+s'\right)=0.$$

On account of the term in s^3, this equation in its general form does not, it would appear, give rise to transformations of much elegance. If, however, $l=n$, then the relation becomes

$$(k-1)s^2+kss'=k^2(m^2-l^2);$$

and in particular, if $k=2$, then

$$s^2+2ss'=4(l^2-m^2),\quad\text{or say}\quad(s+s')^2-s'^2=4(l^2-m^2),$$

viz. taking A instead of b as the fixed point, the relation between the radii AC, Ah is $\rho^2-\rho'^2=4(l^2-m^2)$; the cell is in this case Sylvester's "quadratic-binomial extractor."

618. On the Mechanical Description of a Cartesian

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 328–330.]

[p. 535] Suppose that in two different curves the radius vectors r, r', which belong to the same angle θ, are connected by the equation

$$r^2 + \left(Mr' + N + \frac{P}{r'}\right) r + B = 0;$$

then, taking one of the curves to be the circle

$$Mr' + \frac{P}{r'} = A\cos\theta,$$

the other curve is

$$r^2 + (A\cos\theta + N)r + B = 0,$$

viz. this is a Cartesian. It perhaps would not be difficult to contrive a mechanical arrangement to connect the radius vectors in accordance with the foregoing equation; but the required result may be obtained equally well by means of a particular case of the relation in question; viz. taking this to be

$$r^2 + (-r' + N)r + B = 0,$$

then, taking the one curve to be the circle $r' = A\cos\theta$, the other curve is the Cartesian,

$$r^2 + (-A\cos\theta + N)r + B = 0, \quad \text{that is,} \quad r^2 + (A\cos\theta + N)r + B = 0.$$

The relation between the radius vectors may in this case be written

$$r' = N + r + \frac{B}{r},$$

which can be constructed mechanically by a simple addition to the Peaucellier-cell, viz. if we joint on to C (fig. 1, p. 536) a rod CD having a slot, working on a pin at A, so that the rod is thereby kept always in the line BAC, then, making B the

[p. 536] fixed point and taking $BA = r$, we have $AC = \dfrac{B}{r}$, whence $BC = r + \dfrac{B}{r}$, or D being a point at the distance $CD = a$, from the point C, and denoting BD by r', we have

$$r' = r + \frac{B}{r} + a,$$

which is an equation of the required form; whence, if the point A describe a circle passing through B, then the point D will describe a Cartesian.

Fig. 1.

The equation of the Cartesian is $r^2 + (A\cos\theta + N)r + B = 0$, viz. this is

$$x^2 + y^2 + Ax + B = -N\sqrt{(x^2 + y^2)},$$

or writing $N^2 = a$, it is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ay^2,$$

which is the form considered in the preceding paper. It may be further observed in regard to it that, starting from the focal equation $r = ls + m$, where r, s are the distances of a point

(x, y) of the Cartesian from any two of its three foci, this equation gives $r^2 - l^2 s^2 = m^2 + 2mr$, or writing $r^2 = x^2 + y^2$, $s^2 = (x - a)^2 + y^2$, the function on the left-hand is of the form $(1 + l^2)(x^2 + y^2 + Ax + B)$, whence, assuming $\sqrt{(1 + l^2)} = \sqrt{(a)}$, the equation becomes as above

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2).$$

Taking the distance $r = \sqrt{(x^2 + y^2)}$ to be measured from a given focus, it is easy to see that, no matter which of the other two foci we associate with it, we obtain the same equation $(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2)$; viz. starting with any one focus, we connect with it a determinate circle $x^2 + y^2 + Ax + B = 0$, and a determinate coefficient a, such that taking this focus as the origin, the equation of the curve is

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2);$$

but there are for the given curve three such forms of equation, according as the origin is taken at one or other of the three foci.

(Addition, Feb. 1875.) It is obviously the same thing, but I find that it is mechanically more convenient to derive the Cartesian from the Limaçon $r' = N - A\cos\theta$, by the transformation $r' = r + \frac{B}{r}$: I have on this principle constructed an apparatus whereby the Cartesian is described on a rotating board by a pencil moving in a fixed line.

619. On an Algebraical Operation

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 369–375.]

[p. 537] I consider

$$\Omega F(a, x),$$

an operation Ω performed upon $F(a, x)$ a rational function of (a, x); viz. F being first expanded or regarded as expanded in ascending powers of a, the coefficients of the several powers are then to be expanded or regarded as expanded in ascending powers of x, and the operation consists in the rejection of all negative powers of x.

In the cases intended to be considered, F contains only positive powers of a: but this restriction is not necessary to the theory.

The investigation has reference to the functions $A(x)$ of my "Ninth Memoir on Quantics," *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462]; for instance, we there have as regards the covariants of a quadric

$$A(x) = \frac{1 - x^{-4}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}},$$

and consequently, in the present notation,

$$A(x) = \Omega\, \frac{1 - x^{-4}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}};$$

by a process of development and summation, the value of this expression was found to be

$$\frac{1}{1 - ax^2 \,.\, 1 - a^2},$$

[p. 538] and in the other more complicated cases the value of $A(x)$ was found only by trial and verification. What I purpose now to show is that the operation Ω can be performed without any development in an infinite series; or say that it depends on finite algebraical operations only.

It is clear that if $F(a,x)$, considered to be developed as above contains only positive powers of x, then

$$\Omega F(a,x) = F(a,x);$$

And if it contains only negative powers of x, then $\Omega F(a,x) = 0$.

Consider now $\Omega\,\dfrac{\phi(x)}{x-a}$, where $\phi(x)$ is a function containing only positive powers of x; we have

$$\frac{\phi(x)}{x-a} = \frac{\phi(x)-\phi(a)}{x-a} + \frac{\phi(a)}{x-a},$$

and thence

$$\Omega\,\frac{\phi(x)}{x-a} = \Omega\,\frac{\phi(x)-\phi(a)}{x-a} + \Omega\,\frac{\phi(a)}{x-a} = \frac{\phi(x)-\phi(a)}{x-a},$$

since $\dfrac{\phi(x)-\phi(a)}{x-a}$ is a rational and integral function of x, which when developed contains only positive powers of x, and $\dfrac{\phi(a)}{x-a}$ when developed contains only negative powers of x.

Consider next $\Omega\,\dfrac{\phi(x)}{x^2-a}$, where $\phi(x)$ is a rational and integral function of x; writing this $= f(x^2) + xg(x^2)$, we have

$$\Omega\,\frac{\phi(x)}{x^2-a} = \Omega\,\frac{f(x^2)}{x^2-a} + \Omega\,\frac{xg(x^2)}{x^2-a} = \frac{f(x^2)-f(a)}{x^2-a} + \Omega\,\frac{xg(x^2)}{x^2-a}.$$

As regards the last term, notice that

$$\frac{xg(x^2)}{x^2-a} = \frac{x[g(x^2)-g(a)]}{x^2-a} + \frac{xg(a)}{x^2-a},$$

in which $\dfrac{x[g(x^2)-g(a)]}{x^2-a}$ is a rational and integral function of (a,x), and therefore when developed contains only positive powers of x, while $\dfrac{xg(a)}{x^2-a}$ when developed contains only negative powers of x.

We thus have

$$\Omega\,\frac{\phi(x)}{x^2-a} = \frac{f(x^2)+xg(x^2)-f(a)-xg(a)}{x^2-a} = \frac{\phi(x)-f(a)-xg(a)}{x^2-a}.$$

[p. 539] Similarly, if $\phi(x) = f(x^3) + xg(x^3) + x^2h(x^3)$, then

$$\Omega\,\frac{\phi(x)}{x^3-a} = \frac{\phi(x)-f(a)-xg(a)-x^2h(a)}{x^3-a},$$

and so on.

Consider now the above-mentioned function

$$A(x) = \Omega\,\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}}.$$

Writing

$$\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{P}{1-ax^2} + \frac{Q}{1-a} + \frac{R}{1-ax^{-2}},$$

we have

$$P = \left(\frac{1-x^{-4}}{1-a\,.\,1-ax^{-2}}\right)_{a=x^{-2}} = \frac{1-x^{-4}}{1-x^{-2}\,.\,1-x^{-4}} = \frac{-x^2}{1-x^2},$$

$$Q = \left(\frac{1-x^{-4}}{1-ax^2\,.\,1-ax^{-2}}\right)_{a=1} = \frac{1-x^{-4}}{1-x^2\,.\,1-x^{-2}} = \frac{1}{1-x^2},$$

$$R = \left(\frac{1-x^{-4}}{1-ax^2\,.\,1-a}\right)_{a=x^2} = \frac{1-x^{-4}}{1-x^4\,.\,1-x^2} = \frac{-1}{x^2(1-x^2)};$$

that is,

$$\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{-x^2}{1-x^2}\cdot\frac{1}{1-ax^2} + \frac{1}{1-x^2}\cdot\frac{1}{1-a} + \frac{-1}{x^2(1-x^2)}\cdot\frac{1}{1-ax^{-2}},$$

and thence

$$\Omega\,\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{-x^2}{1-x^2}\cdot\frac{1}{1-ax^2} + \frac{1}{1-x^2}\cdot\frac{1}{1-a} + \Omega\,\frac{-1}{x^2(1-x^2)}\cdot\frac{1}{1-ax^{-2}}.$$

Here, as regards the last term,

$$\Omega\,\frac{1}{1-ax^{-2}} = \Omega\,\frac{-x^2}{x^2-a} = \Omega\,\frac{\phi(x^2)}{x^2-a} = \frac{\phi(x^2)-f(a)}{x^2-a} = \frac{-x^2+a}{x^2-a},$$

which gives

$$\Omega\,\frac{-1}{x^2(1-x^2)}\cdot\frac{1}{1-ax^{-2}} = \frac{1}{1-x^2}\cdot\frac{x^2+a}{1-ax^2}\cdot\frac{1}{x^2}\cdot ?$$

and we have

$$A(x) = \frac{-x^2}{1-x^2}\cdot\frac{1}{1-ax^2} + \frac{1}{1-x^2}\cdot\frac{1}{1-a} + \frac{1}{1-x^2}\cdot\frac{x^2+a}{1-ax^2\,.\,1-a}.$$

The second term is $= \dfrac{1}{1-x^2}\cdot\dfrac{1}{1-a}$: combining this with the third term, the two together are $= \dfrac{1+ax^2}{1-x^2\,.\,1-ax^2}$.

[p. 540] Hence the value is

$$= \frac{1}{1-x^2}\left(\frac{-x^2}{1-ax^2} + \frac{1+ax^2}{1-ax^2}\right),$$

which is

$$= \frac{1}{1-ax^2\,.\,1-a^2},$$

being, in fact, the expression for this function when decomposed into partial fractions of the denominators $1-ax^2$ and $1-a^2$ respectively. Hence finally

$$\Omega\,\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{1}{1-ax^2\,.\,1-a^2},$$

as it should be.

For the cubic function, we have

$$A(x) = \Omega\,\frac{1-x^{-6}}{1-ax^3\,.\,1-ax\,.\,1-ax^{-1}\,.\,1-ax^{-3}};$$

the function operated upon, when decomposed into partial fractions, is

$$= \frac{x^6}{1-x^2\,.\,1-x^4\,.\,1-ax^3} - \frac{x^2}{1-x^2\,.\,1-x^6\,.\,1-ax}$$

$$+\frac{x}{1-x^2\,.\,1-x^6}\cdot\frac{1}{x-a}-\frac{x}{1-x^2\,.\,1-x^4}\cdot\frac{1}{x^3-a}.$$

Hence we require

$$\Omega\,\frac{1}{1-x^2\,.\,1-x^6}\cdot\frac{x}{x-a}+\Omega\,\frac{1}{1-x^2\,.\,1-x^4}\cdot\frac{-x}{x^3-a}.$$

The first of these is

$$=\frac{1}{x-a}\left(\frac{x}{1-x^2\,.\,1-x^6}-\frac{a}{1-a^2\,.\,1-a^6}\right),$$

which is

$$\frac{1}{1-x^2\,.\,1-x^4\,.\,1-a^2\,.\,1-a^6}\left\{\begin{array}{l} 1 \\ +x(a+a^3-a^5) \\ +x^2(a^2+a^4) \\ +x^3(a-a^5) \\ -x^4a^5 \\ -x^6a \end{array}\right\}.$$

As regards the second, the function operated on may be expressed in the form

$$\frac{-x-x^3-x^5}{1-x^2\,.\,1-x^6}\cdot\frac{1}{x^3-a},$$